

Forecasting fine-particle (PM2.5) pollution in central Paris one day ahead

Professional report — RNCP37827 « Développeur en intelligence artificielle » (level 6), block BC01: « Réaliser la collecte, le stockage et la mise à disposition des données d'un projet en intelligence artificielle »

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GitHub repository: <https://github.com/JujuGnirt/air-quality-paris>

1. Use case, goal and objectives

Health stakes. Fine particles (PM2.5, diameter below 2.5 μm) are the air pollutant with the largest health impact: they reach deep into the lungs and the bloodstream and cause cardiovascular and respiratory disease. In France, they are linked to nearly **40,000 deaths a year**, 7 % of total mortality [1]; in Île-de-France, 6,220 deaths in 2019 were attributable to long-term PM2.5 exposure [2]. The World Health Organization (WHO) states that health effects occur at any level and in 2021 lowered its daily guideline to **15 $\mu\text{g}/\text{m}^3$** [3]. French information and alert thresholds are far less strict, so many moderately polluted days go unnoticed by the public.

Problem statement. Paris has improved over the last decade, yet about one day in five still exceeds the WHO guideline in the city centre. Residents cannot change the air, but they can limit their exposure on bad days if they know in advance. Hence the question: *can we forecast, on the evening of day D, whether PM2.5 in central Paris will exceed the WHO guideline on day D+1?* A reliable day-ahead signal would let sensitive people (asthmatics, children, elderly people) and local services adapt their activities.

Objectives.

1. Collect air-quality, weather and calendar data from five types of sources (web scraping, API, flat files, relational database, big-data system).
2. Clean, document and store them in a normalised MySQL database, expose them through a REST API (Flask) and load a denormalised copy into BigQuery.
3. Explore the data (EDA) to understand what drives PM2.5.
4. Build and evaluate a machine-learning model forecasting D+1 PM2.5 and WHO exceedances, and compare it with a simple baseline.

Scope and specifications.

Item	Content
Actors	Data providers: Airparif, Météo-France, api.gouv.fr , Ministry of Education. Users: sensitive people and local services (future alert); technical teams (analysts, data scientists) consuming the API. Author: collection, storage, API and model.
Functional objectives	Tell the evening before whether PM2.5 will exceed 15 $\mu\text{g}/\text{m}^3$ in central Paris; make the cleaned data available to other components through an API.
Technical objectives	Re-runnable collection scripts for 5 source types; cleaned, normalised MySQL database; REST API protected by a key and documented (OpenAPI); denormalised BigQuery table; evaluated ML model.
Environment and tools	Windows laptop, Anaconda (Python 3, Jupyter), MySQL 8 Community + Workbench (local), Google BigQuery (cloud), Git/GitHub, Trello. Python libraries: pandas, requests, BeautifulSoup, pdfplumber, SQLAlchemy, PyMySQL, Flask, scikit-learn, matplotlib (<code>requirements.txt</code>).
External services	Météo-France API (free account, API key, asynchronous orders); public-holiday API (no key); BigQuery (Google Cloud project).
Constraints	No personal data; secrets outside the code; open licences (ODbL, Licence Ouverte); BigQuery free tier; API served locally (127.0.0.1), available when the author runs it.
Budget	0 € of spending: open data, free API account, BigQuery free tier, free software. Human budget: one person over the project period planned in Trello (section 2).

2. Project planning

The project was planned in a Trello board (columns *To Do* / *In Progress* / *Done*, one card per deliverable, due dates on the critical items) and organised in five phases:

Phase	Content	Deliverables
1. Scoping	Use case, problem statement, choice and check of the data sources	Problem statement, source list
2. Collection & cleaning	Airparif flat files, web scraping of episodes, Météo-France API, holidays; cleaning pipeline	src/ scripts, notebook 01, data dictionary
3. Storage & exposure	MySQL schema + loading, ERD, 5 SQL queries, Flask API, BigQuery	sql/ , api_flask.py , BigQuery table
4. Analysis	EDA, feature engineering, models, evaluation	Notebooks 02 and 03, figures
5. Communication	Report, slides, GitHub repository, rehearsal	This report, slides, repository

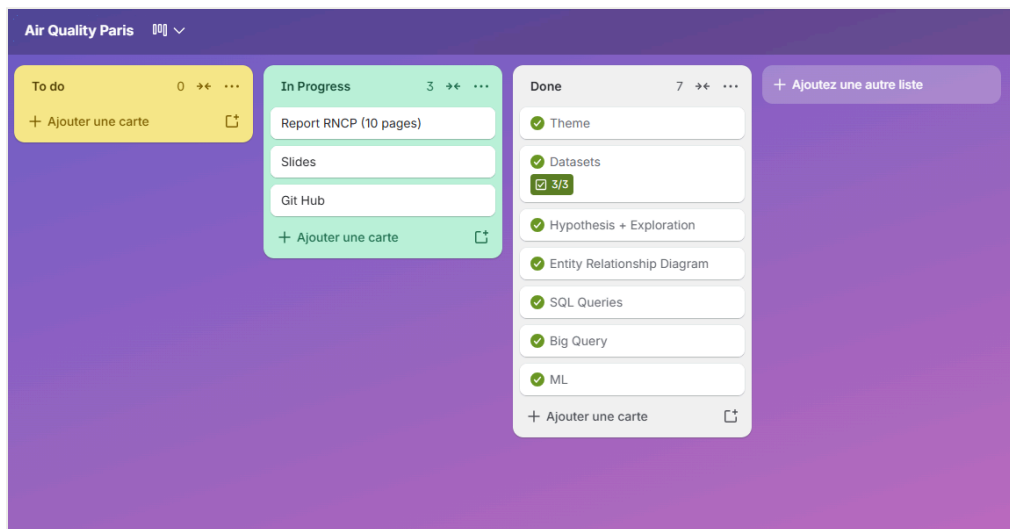


Figure 0 — Trello board of the project.

3. Data sources

Every required source type is used; each one answers a precise need of the model.

Type	Source	Data	Why this source
Flat files (CSV)	Airparif exports (official air-quality monitoring body of the Paris region); Airparif Open Data <code>airt_idf.csv</code> ; school calendar (<code>data.education.gouv.fr</code>)	Hourly PM2.5, NO2, NOx, O3, CO, PM10 at 52 stations, 2018 → 21/09/2026 (9.3 million values); alert procedures 2019-2025; school holidays	The target variable and the explanatory pollutants; only official, accredited measurements are acceptable for health questions
Web scraping	<code>airparif.fr</code> : HTML table of the current year + yearly PDF reports ("Bilan des épisodes")	Official pollution episodes (date, pollutant, level, forecast / observed, forecast quality)	No downloadable file exists; the PDFs also say whether Airparif had forecast each episode — an external reference for our model
API	Météo-France "Données Publiques Climatologie" (with key); <code>calendrier.api.gouv.fr</code>	Daily observed weather at Paris-Montsouris (temperature, wind, rain, pressure, humidity), 3,186 days; public holidays	Weather is the main driver of day-to-day pollution; Météo-France is the official source
Relational database	MySQL 8 (loaded from Python with SQLAlchemy)	Normalised star schema, 6 tables	Integrity (keys, constraints), SQL analysis, back-end of the API
Big-data system	Google BigQuery	Denormalised table, 386,561 rows × 27 columns, partitioned and clustered	Public datasets were checked first: the OpenAQ table stops in May 2022 with only 828 PM2.5 values for Paris, so our own data were loaded

Collection details worth noting. Scraping respects `robots.txt`, identifies itself with an explicit User-Agent and pauses 2 s between requests. The Airparif PDF tables could not be read with the standard text extraction (characters flagged as vertical text, subscript digits on another line): lines were rebuilt from the position of each character with `pdfplumber`. The Météo-France API is asynchronous (order → HTTP 202, then poll the file until HTTP 201), one order per year. The API key is stored in `secrets/keys.env`, excluded from Git.

4. Metadata and data dictionary

Final analysis table `dataset_daily.csv` (one row per day, 01/01/2018 → 21/09/2026, 3,186 rows, 27 columns). Pollutant values are daily means at the central-Paris urban background station (PA04C until 22/09/2019, then PA01H, 1 km apart).

Column(s)	Type	Unit	Source	Description
<code>date</code>	date	—	—	Day (Europe/Paris time)
<code>pm25</code>	float	µg/m ³	Airparif CSV	Target: daily mean PM2.5 (valid if ≥ 18 hours out of 24)
<code>no2</code> , <code>nox</code> , <code>o3</code> , <code>pm10</code> / <code>co</code>	float	µg/m ³ / mg/m ³	Airparif CSV	Other pollutants, same station; <code>o3_max</code> = highest hourly ozone
<code>source_station</code>	text	—	derived	Station that measured PM2.5 that day (PA04C / PA01H)
<code>temp_mean_c</code> , <code>temp_min_c</code> , <code>temp_max_c</code>	float	°C	Météo-France API	Mean, minimum, maximum temperature (Météo-France codes TM, TN, TX)
<code>humidity_mean_pct</code>	int	%	Météo-France API	Mean relative humidity (UM)
<code>pressure_mean_hpa</code>	float	hPa	Météo-France API	Mean sea-level pressure (PMERM)
<code>rain_mm</code>	float	mm	Météo-France API	Daily precipitation (RR)
<code>wind_mean_ms</code> , <code>wind_max_ms</code> , <code>wind_dir_deg</code>	float	m/s, °	Météo-France API	Mean wind (FFM), max 10-min wind (FXY) and its direction (DXY)
<code>day_of_week</code> , <code>month</code> , <code>is_weekend</code>	int	—	derived	Calendar (0 = Monday)
<code>is_public_holiday</code> / <code>is_school_holiday</code>	0/1	—	API gouv / CSV	Public holiday / school holiday (zone C, Paris)
<code>episode_pm10</code> , <code>episode_o3</code> , <code>episode_no2</code>	0/1	—	Scraping + Open Data	Official information or alert procedure (or observed exceedance) that day
<code>above_who</code>	0/1	—	derived	1 if <code>pm25</code> > 15 µg/m ³ (WHO 2021 daily guideline)

Full descriptions of every intermediate table (stations, pollutants, hourly and daily measurements, episodes, ring road, 2014-2017 indices) are in the repository README.

5. Data cleaning

Issue found	Decision	Justification
Wide format: 6 header rows, one column per station	Headers read separately (metadata); <code>melt</code> to long format	One row = one measurement, suited to SQL and pandas
Same station code twice in a file (RD934)	Columns identified by position	Avoids mixing a full and an empty column
Timestamps in UTC, marking the end of the hour	-1 h, then conversion to Paris time	Days aligned with Parisian life, daylight saving included
3,113 negative values (0.03 %, minimum -3 µg/m³)	Set to 0	Device noise in clean air; deleting them would bias means upwards
965 duplicates (first hour of the next year in each file)	Dropped on (hour, station, pollutant)	Unique key of a measurement
Station type not given	Inferred from the name with whole-word regex (traffic / urban background / rural)	Needed to compare traffic and background; "Champigny-sur-Marne" was first misclassified as a road
Incomplete days	Daily mean kept only if ≥ 18 hours measured (7,177 of 393,738 days rejected)	Usual 75 % completeness rule
Station change in 2019 (PA04C → PA01H)	Joined into one "central Paris" series, origin kept in <code>source_station</code>	Same station type, 1 km apart, never measured on the same day; assumption reported
Episodes from 3 sources (PDF, HTML, Open Data)	Merged; priority PDF > HTML > Open Data	Keep the most complete row (forecast + observed)
Missing days	Complete calendar; gaps left as NaN, never imputed	Makes gaps visible; imputing the target would fake the evaluation

The 2014-2017 files are public-information *indices* per municipality, not concentrations; Airparif advises against using them for trend analysis, so they are stored in a separate context table and never joined to measurements.

6. Relational database (MySQL)

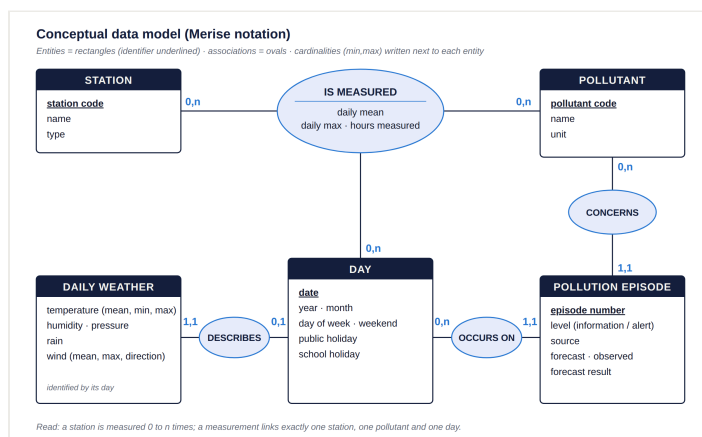


Figure 1a — Conceptual data model (Merise): business entities, associations and cardinalities, no technical types.

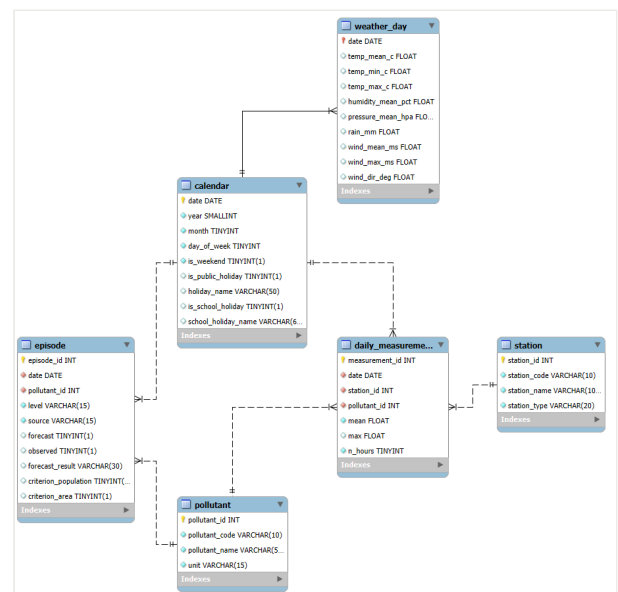


Figure 1b — Physical model (MySQL Workbench, reverse engineering): 6 tables, 6 foreign keys.

From conceptual to physical model. The ternary association *IS MEASURED* (station × pollutant × day) becomes the fact table `daily_measurement`, whose foreign keys are the identifiers of the three entities; each 1,1 cardinality becomes a foreign key on that side (`episode.pollutant_id`, `episode.date`, `weather_day.date`); business codes are replaced by numeric surrogate keys created in Python.

Star schema. The fact table `daily_measurement` (386,561 rows: one daily mean per station × pollutant × day) points to three dimensions — `station` (52), `pollutant` (6), `calendar` (3,186 days) — and two context tables are linked to the calendar: `weather_day` (1:1) and `episode` (107 rows, N:1 to calendar and pollutant).

- **Normalisation:** station names and units stored once; the fact table only holds numeric keys created in Python.
- **Why MySQL:** the data are tabular and strongly linked (every measurement refers to a station, a pollutant and a day), so a relational database enforces these links with foreign keys and answers the analysis questions with joins; MySQL is free, runs locally and is the system used in the training. A document database (NoSQL) would not check the links, and a warehouse like BigQuery is kept for the big-data copy.
- **Integrity:** primary and foreign keys, uniqueness on (date, station, pollutant), index on (station, pollutant).
- **Loading with pandas + SQLAlchemy** (to `sql`, if_exists="append" into tables created by `schema.sql`, so keys are kept); dimensions first, then facts; row counts checked (6/6 OK).
- The 9.3 million hourly values are not stored in MySQL; the big-data copy goes to BigQuery.

Five analysis queries (`sql/analysis_queries.sql`)

#	Question	SQL techniques	Main result
1	Is PM2.5 going down?	4-table joins, GROUP BY	Urban background: 12.8 µg/m³ (2018) → 8.4 (2024); traffic stations always higher
2	Which months are most at risk?	Conditional aggregation	March 37.5 %, January 34.5 % of days above the WHO guideline; August 5 %
3	How much NO2 does traffic add?	CTE + CASE pivot	Traffic ≈ 2 × background; the excess fell from 30.8 to ≈ 15 µg/m³
4	What weather comes with the worst days?	LEFT JOIN, ORDER BY / LIMIT	Top 10 all Dec-Mar, ≤ 0.2 mm rain, pressure > 1,020 hPa, wind < 2.5 m/s for 8 of 10
5	Do official episodes show in our data?	Window function LAG, CTE	48 of 49 PM10 episode days above the WHO PM2.5 guideline, incl. all 5 "false alarms"

```

105 • SELECT
106     m.date,
107     ROUND(m.mean, 1) AS pm25_ug_m3,
108     w.temp_mean_c,
109     w.wind_mean_ms,
110     w.rain_mm,
111     w.pressure_mean_hpa
112 FROM daily_measurement m
113 JOIN station s ON s.station_id = m.station_id
114 JOIN pollutant p ON p.pollutant_id = m.pollutant_id
115 LEFT JOIN weather_day w ON w.date = m.date
116 WHERE p.pollutant_code = 'PM25'
117 AND s.station_code IN ('PA01H', 'PA04C')
118 ORDER BY m.mean DESC LIMIT 10;

```

date	pm25_ug_m3	temp_mean_c	wind_mean_ms	rain_mm	pressure_mean_hpa
2022-01-15	68.5	2.7	1.5	0.2	1027.5
2023-02-15	57.5	7	1.7	0	1027.1
2018-02-21	54.6	2.6	4.5	0	1022.3
2020-01-25	52	3.1	1.3	0	1020.5
2019-02-22	50	11	1.6	0	1034.6

Figure 2 — Query 4 in MySQL Workbench: the ten most polluted days and their weather.

7. REST API (Flask)

The API (`src/api_flask.py`) reads MySQL and returns JSON. **Two resources, four endpoints**, each resource having a paginated list with filters and a single-object endpoint with nested details:

Endpoint	Returns	Parameters
GET <code>/stations</code>	Paginated list of stations	<code>type</code> , <code>name</code> , <code>page</code> , <code>per_page</code>
GET <code>/stations/<code></code>	One station + summary per pollutant (days measured, period, mean, days above WHO) and links	—
GET <code>/measurements</code>	Paginated daily measurements, most recent first	<code>station</code> , <code>pollutant</code> , <code>start</code> , <code>end</code> , <code>min_value</code> , <code>page</code> , <code>per_page</code>
GET <code>/measurements/<id></code>	One measurement with nested station, pollutant, weather of the day and episodes of the day	—

Design choices: pagination (20 by default, 100 maximum) with `total_items`, `total_pages` and next/previous links; filter values passed as SQL parameters (protection against SQL injection); JSON errors (400 invalid parameter, 401 missing or invalid key, 404 not found); an HTML documentation page at `/` and an **OpenAPI 3 specification** (`docs/openapi.yaml`, also served at `/openapi.yaml`).

Authentication. Every data endpoint requires an API key sent in the HTTP header `X-API-Key`; the valid key is read from `secrets/keys.env` (never in the code or on GitHub). Without a valid key the API answers `401`; the documentation pages stay public because they contain no data.

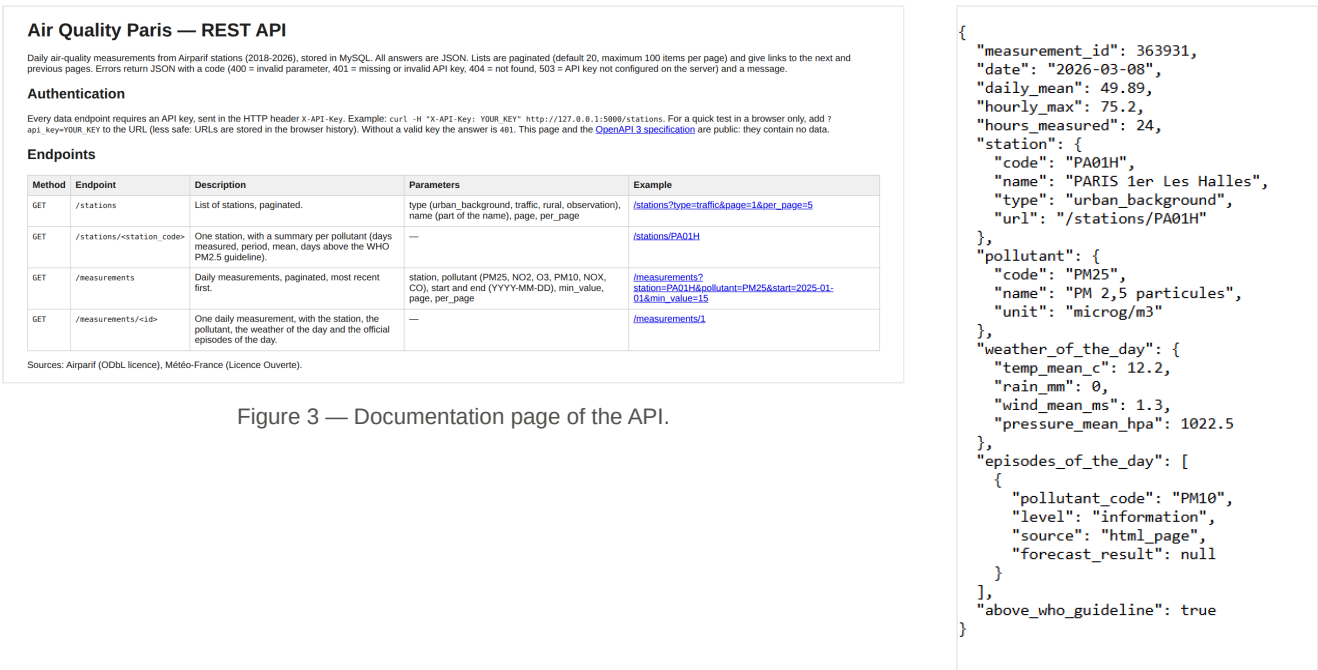


Figure 3 — Documentation page of the API.

Figure 4 — `/measurements/363931`: 8 March 2026, 49.9 µg/m³, calm anticyclonic weather and an official PM10 episode.

8. Big-data system (BigQuery)

Public datasets first. The BigQuery public dataset `bigquery-public-data.openair.global_air_quality` was queried for Paris: it holds only 828 PM2.5 values, from April 2018 to **26 May 2022**, when it stops being updated. It covers neither 2023-2026 nor a daily series per station, so a **denormalised version of our database** was loaded instead (`src/export_bigquery.py`).

- **Denormalised table** `air_quality_paris.daily_measurements`: one row per station × pollutant × day with station, pollutant, calendar, weather and episode columns (386,561 rows × 27 columns). In a warehouse,

storage is cheap and joins on large tables are expensive, so the data are stored "ready to query".

- **Partitioning** by integer range on `year` (2018 → 2027, step 1) rather than by date: it matches our queries, which filter by year, and it stays compatible with the free sandbox, which deletes date partitions older than 60 days and would have erased the history.
- **Clustering** on `pollutant_code`, `station_code`, the two most frequent filters.

Check (`INFORMATION_SCHEMA.PARTITIONS`): 9 partitions, 2018 to 2026, whose row counts add up to exactly 386,561 — no year was lost (2021 is smaller because there is no NO2 file that year).

Measured effect (bytes processed, from the job information of each query):

Query	Filter	Bytes processed	What reduced the scan
Mean of all rows	none	2.95 MB	— (full scan of the column)
Mean for 2024	<code>year = 2024</code>	0.64 MB	Partition pruning: ÷ 4.6
Monthly PM2.5 at PA01H, 2024 (query 1a)	<code>year + pollutant + station</code>	1.42 MB	Clustering: the same 1.42 MB whether the year is filtered through the partition column or through <code>date</code>
Same with a <code>date</code> filter (query 1b)	<code>date + pollutant + station</code>	1.42 MB	
NO2 traffic vs background, all years (query 2)	<code>pollutant only</code>	12.85 MB	—

Résultats de la requête	
Informations sur le job	Résultats
Emplacement	EU
Date et heure de création	27 sept. 2026, 21:56:46 UTC+2
Heure de début	27 sept. 2026, 21:56:46 UTC+2
Heure de fin	27 sept. 2026, 21:56:46 UTC+2
Durée	166 ms
Octets traités	635 48 ko
Octets facturés	10 Mo
Millisecondes d'emplacement	24
Priorité de la tâche	INTERACTIVE
Utiliser l'ancien SQL	false
Table de destination	Table temporaire
Mode d'utilisation de l'index	UNUSED
Motifs de non utilisation de l'indexation	There is no index configuration for the base table 'project-5b3d678-14f9-4b0f-a00-air_quality_paris.daily_measurements'.
Étiquettes	
1	<code>SELECT AVG(daily_mean) FROM air_quality_paris.daily_measurements WHERE year = 2024;</code>

Figure 5 — Query filtered on the partition column: 635 KB processed.

Résultats de la requête	
Informations sur le job	Résultats
Emplacement	EU
Date et heure de création	27 sept. 2026, 21:57:24 UTC+2
Heure de début	27 sept. 2026, 21:57:24 UTC+2
Heure de fin	27 sept. 2026, 21:57:24 UTC+2
Durée	195 ms
Octets traités	2,95 Mo
Octets facturés	10 Mo
Millisecondes d'emplacement	59
Priorité de la tâche	INTERACTIVE
Utiliser l'ancien SQL	false
Table de destination	Table temporaire
Étiquettes	
1	<code>SELECT AVG(daily_mean) FROM air_quality_paris.daily_measurements;</code>

Figure 6 — Same query without filter: 2.95 MB processed.

Reading: when a query filters on station and pollutant, clustering already skips most blocks, so the partition adds nothing; when it filters only on time, the partition does the work. At this volume (58 MB) the cost is negligible either way (BigQuery bills at least 10 MB per query), but the same design scales to the 9.3 million hourly values.

9. Exploratory data analysis

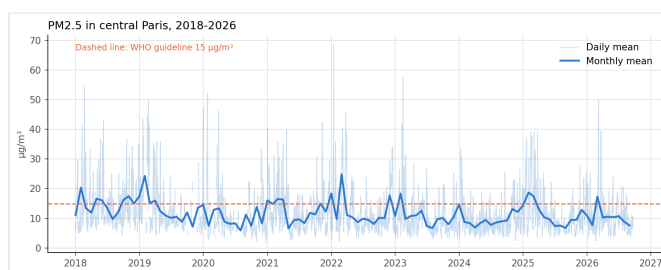


Figure 7 — Daily and monthly PM2.5 in central Paris.

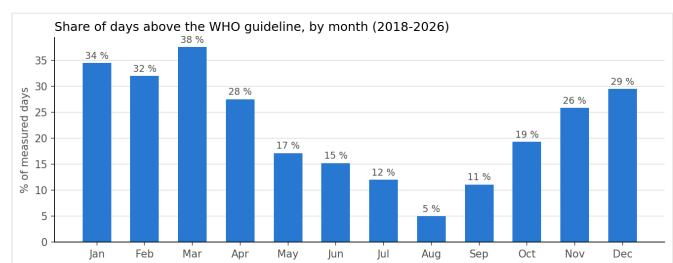


Figure 8 — Share of days above the WHO guideline by month.

- **Downward trend:** yearly mean 14.5 µg/m³ in 2018 → 9.7 in 2024 (11.6 in 2025), but 14 % to 34 % of days still exceed the WHO guideline: the problem is the peaks.
- **Strong seasonality:** January-March concentrate exceedances (heating, stable cold air, spring agricultural ammonia forming secondary particles); August only 5 %. Weekday effects are small (≈ 1 µg/m³).

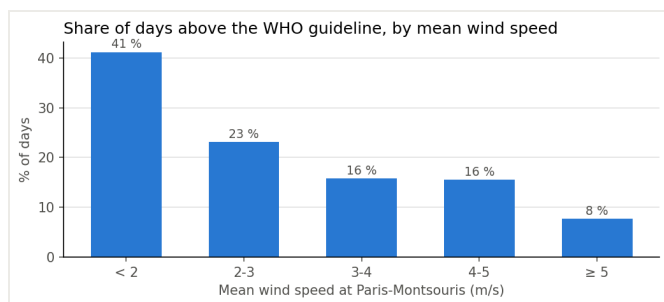


Figure 9 — Exceedances by mean wind speed.

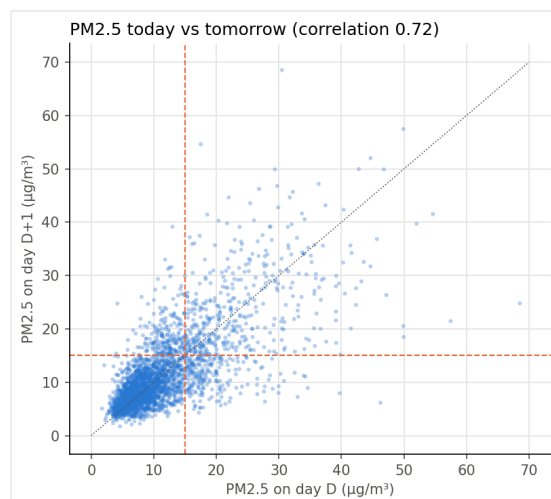


Figure 10 — PM2.5 today vs tomorrow.

- **Weather:** calm days (< 2 m/s) exceed the guideline in 41 % of cases, windy days (≥ 5 m/s) in 8 %; cold and high pressure also go with pollution (correlations: pressure +0.31, temperature −0.25, wind −0.28 to −0.31). NO_x, NO₂ and CO rise with PM_{2.5} (+0.57 to +0.65).
- **Persistence:** if today exceeds 15 µg/m³, tomorrow does too in 65 % of cases (10 % otherwise) → lagged features and a persistence baseline.
- **Validation of the sources:** on official PM₁₀ episode days, mean PM_{2.5} is 35.0 µg/m³ against 11.3 on other days; 98 % of them exceed the WHO guideline.

10. Machine learning: D+1 forecast

Process

- **Target:** PM_{2.5} of D+1 (regression); an alert is raised when the forecast exceeds a threshold, evaluated as a classification of WHO exceedances.
- **Feature engineering** (only information available on the evening of D): PM_{2.5} of D, D−1, D−2 and 7-day mean; NO_x and ozone peak of D; weather of D+1 (temperature, wind speed and direction, rain, pressure, humidity, pressure change); month, day of week and holidays of D+1 — 16 features, 2,531 usable days. PM₁₀, CO and NO₂ were excluded (too many gaps) and official episodes too (announced the same day: leakage).
Assumption: observed D+1 weather stands in for the weather forecast.
- **Time-based split** (no random shuffling of a time series): train 2018-2023 (1,701 days), validation 2024 (296), test 2025 (280), demo 2026 (254).
- **Models:** persistence baseline ("tomorrow = today"), linear regression, Random Forest.
- **Hyperparameter tuning:** grid on max_depth {None, 8, 12}, min_samples_leaf {1, 5, 10}, max_features {1.0, 0.5}, selected on validation MAE (best: 12 / 5 / 0.5, MAE 2.66). A fixed validation year replaces k-fold cross-validation, whose folds would mix past and future.
- **Imbalance** (≈ 20 % exceedance days): accuracy is not used; instead MAE/RMSE and recall/precision/F1 on exceedances. Rather than re-sampling, the **alert threshold** is tuned on 2024: the highest threshold keeping recall ≥ 80 % (a missed exceedance costs more than a false alarm) → **13 µg/m³**.

Results on the unseen test year (2025), models retrained on 2018-2024

Model	MAE (µg/m ³)	RMSE	R ²	Recall	Precision	F1
Persistence (baseline)	3.88	5.66	0.41	0.67	0.67	0.67
Linear regression	3.21	4.69	0.59	0.81	0.72	0.76
Random Forest, alert at 15	2.99	4.50	0.63	0.78	0.74	0.76
Random Forest, alert at 13	2.99	4.50	0.63	0.94	0.57	0.71

-23 %

error vs persistence
(MAE 2.99 vs 3.88)

60 / 64

exceedances
announced the day
before in 2025

46

false alarms in 2025
(216 clean days)

2.73

MAE on 2026
(persistence 3.34); recall
0.76

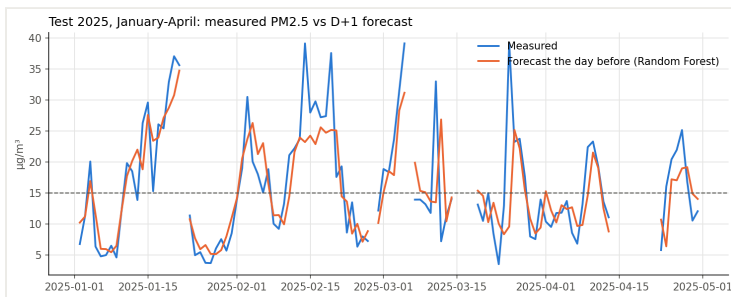


Figure 11 — Test 2025, January-April: measured vs forecast the day before.

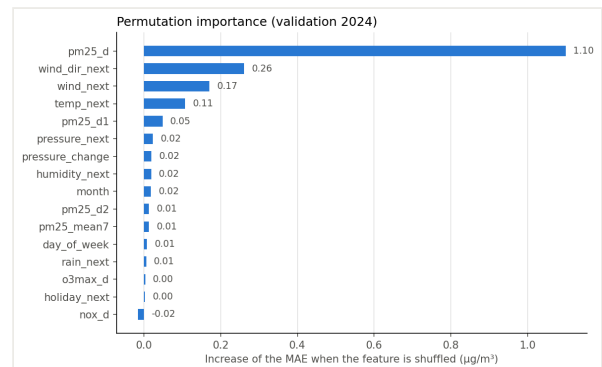


Figure 12 — Permutation importance (validation 2024).

Interpretation. Today's PM2.5 is by far the most important feature, followed by tomorrow's wind (direction first — north-easterly winds bring continental air — then speed) and temperature; NOx adds nothing once PM2.5 is known. The model follows episodes well but smooths the highest peaks and reacts one day late to sudden starts driven by factors it does not see (long-range transport, agricultural spreading). In 2026 recall drops to 0.76: performance varies from one year to the next.

11. Conclusions, limitations and next steps

Conclusions. PM2.5 in Paris is decreasing, but WHO exceedances remain frequent in winter and early spring and are driven by calm, cold, anticyclonic weather and by persistence from one day to the next. A Random Forest using today's pollution and tomorrow's weather forecasts D+1 PM2.5 with a mean error of about 3 µg/m³ and announces nine exceedances out of ten in 2025, clearly beating the persistence baseline.

Limitations. Observed weather used as a stand-in for the forecast (real performance slightly lower); a single urban background station (not valid next to heavy traffic); station change in 2019; peaks under-estimated; about one false alarm per correct alert at the chosen threshold; no NO₂ file for 2021; ring-road files probably modelled rather than measured (to be confirmed with Airparif).

Next steps. Use real Météo-France forecasts; add boundary-layer height and regional background stations (imported pollution); try gradient boosting and D+2/D+3 horizons; serve the forecast through a `/forecast` endpoint of the API; schedule the collection.

12. GDPR, ethics and licences

- **No personal data in the database:** environmental measurements, weather, calendars and official public episodes only. The **register of processing activities** and the **data-sorting procedures** are in `docs/rgpd.md`: 8 processing activities listed; the only personal data met is the IP address shown in the Flask console (not stored); 6 procedures with their frequency (column check at each schema change and monthly, minimisation at loading, log deletion, secrets check before each push, yearly review).
- **Scraping ethics:** public pages only, `robots.txt` respected, identified User-Agent, pauses between requests, raw files kept to avoid re-downloading.
- **Security:** API access protected by a key (section 7); API keys and database password stored in `secrets/keys.env`, excluded by `.gitignore`, never in the code or the repository.
- **Licences:** Airparif data under ODbL (attribution: "Source: Airparif"); Météo-France, `data.gouv.fr` and `data.education.gouv.fr` data under Licence Ouverte / Etalab 2.0.

13. Sources

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- Météo-France — API Données Publiques Climatologie: <https://portail-api.meteofrance.fr/web/fr/api/DonneesPubliquesClimatologie>
- Public holidays API: <https://calendrier.api.gouv.fr> · School calendar: <https://data.education.gouv.fr>
- Google Cloud — BigQuery sandbox documentation: <https://cloud.google.com/bigquery/docs/sandbox>
- Tools: Python (pandas, numpy, requests, BeautifulSoup, pdfplumber, SQLAlchemy, Flask, scikit-learn, matplotlib), MySQL 8 / Workbench, Google BigQuery, Trello.

AI use statement. Claude (Anthropic) was used throughout the project to generate and debug code, suggest methods and draft the report and presentation. The author executed all collection, loading and modelling steps herself, reviewed the code, and verified the results. Several AI outputs were corrected during this review, for example a wrong episode date, an unverified claim about the OpenAQ dataset, and a false-alarm count. The final choices and their justification are the author's.